Keshav Sharma

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EDUCATION

Vellore Institute of Technology **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, Java, C/C++, SQL, JavaScript, R

Data Engineering & Pipelines: Data Collection & Processing, Data Quality & Integrity, ETL Pipeline Design, Time-Series Data Analysis, Feature Engineering, Pandas, NumPy

Machine Learning & Data Science: Machine Learning, Deep Learning (PyTorch), Time-Series Forecasting, Anomaly Detection, Probabilistic Modeling (BG/NBD, Gamma-Gamma), Scikit-learn, Model Training & Evaluation

Full-Stack Development: Flask, FastAPI, React.js, Node.js, RESTful APIs

Databases & Storage: MySQL, SQLite, Redis (NoSQL), SQL Query Optimization

Tools & Platforms: Git, GitHub, GitHub Actions, Docker, PyTest, CI/CD, Power BI

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** ***Aug – Sep, 2025

- Collected and analyzed 2 real-world datasets (8,500+ records) using LLM-based natural-language querying, extracting 15 structured business insights.
- Iteratively validated AI-generated outputs for statistical consistency and accuracy, ensuring findings were reliable before use.
- Designed a repeatable LLM-assisted analytics workflow combining data preprocessing, natural-language querying, insight extraction, and validation to convert raw datasets into actionable business findings.

PROJECTS

***Customer Lifetime Value Engine* |LINK**

- Designed a data pipeline to collect, clean, and process 1M+ historical transaction records, feeding a probabilistic ML model (BG/NBD + Gamma-Gamma) for lifetime value forecasting.
- Ensured data quality through validation and cleaning steps, then engineered features to support model training and evaluation.
- Built Power BI dashboards to visualize trends and communicate data-driven insights to stakeholders.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Built a time-series data pipeline for NSE stock data, applying Isolation Forest and a PyTorch-based LSTM Autoencoder to forecast and flag abnormal price/volume behavior.
- Integrated FinBERT-based sentiment scoring on financial news as an auxiliary data source, combining structured and unstructured data into a single pipeline.
- Achieved ~88% precision and ~85% recall on a labeled validation set; performed feature engineering and model evaluation to optimize forecasting accuracy.

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory data store in Python with a custom RESP protocol parser, multithreaded TCP server, and an O(1) LRU cache with lazy/active TTL expiration — core data pipeline infrastructure for high-throughput storage and retrieval.
- Designed a four-layer architecture (transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL) and automatic crash recovery to ensure data integrity.
- Validated the system with 46 automated PyTest unit tests, benchmarking 619K SET ops/sec in-memory and 2.8K ops/sec with durable WAL persistence.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built a full-stack application using FastAPI, React, SQLAlchemy, and SQLite, automating receipt data extraction and processing via EasyOCR, OpenCV, and NLP-based parsing.
- Engineered RESTful APIs with JWT authentication and Google OAuth, plus a responsive analytics dashboard for budget tracking and spend insights.
- Deployed the application across Render and Vercel, managing frontend-backend integration and production configuration end to end.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Prepare Data for ML APIs on Google Cloud Skill Badge |**LINK** 2026
- AWS APAC Solutions Architecture virtual experience program | **LINK** 2025